PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

# Exam 1 · Version B · Solutions

20 questions · 90 minutes · original practice exam, not an official exam

## ANSWER KEY

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|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **b** | 2 | **b** | 3 | **c** | 4 | **c** | 5 | **e** | 6 | **a** | 7 | **c** |
| 8 | **d** | 9 | **a** | 10 | **c** | 11 | **d** | 12 | **a** | 13 | **c** | 14 | **b** |
| 15 | **b** | 16 | **b** | 17 | **b** | 18 | **b** | 19 | **c** | 20 | **d** | | |

**THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A ball is thrown straight up at 15 m/s from a point 2 m above the ground. Ignore air resistance.

**1)** What is the maximum height of the ball above the ground?

**Answer: (b) 13.5 m**

$h = h_0 + v_0^2/2g = 2 + 11.47 = 13.5$ m.

**2)** How long after it is thrown does the ball hit the ground?

**Answer: (b) 3.19 s**

Solve $0 = 2 + 15t - 4.905t^2$: $t = 3.19$ s.

**3)** At the top of its path, the acceleration of the ball is

**Answer: (c) 9.81 m/s² downward.**

Gravity acts the whole time. Only the velocity is zero at the top.

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**THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A car waits at a red light. The instant the light turns green, the car accelerates at 3 m/s², and at that same instant a truck passes it moving at a constant 20 m/s.

**4)** How long does it take the car to catch the truck?

**Answer: (c) 13.3 s**

$\frac{1}{2}at^2 = v_T t$, so $t = 2v_T/a = 13.3$ s.

**5)** How fast is the car moving when it catches the truck?

**Answer: (e) 40 m/s**

$v = at = (3)(13.3) = 40$ m/s, twice the truck's speed.

**6)** At the instant the car is moving as fast as the truck, the car is

**Answer: (a) behind the truck.**

Until then the truck has been faster, so the gap has been growing. The car is still behind.

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**THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A ball is launched from the roof of a 30 m tall building at 12 m/s, 30° above the horizontal. Ignore air resistance.

**7)** How far from the base of the building does the ball land?

**Answer: (c) 32.8 m**

Solve $0 = 30 + 6t - 4.905t^2$: $t = 3.16$ s. $x = (10.4)(3.16) = 32.8$ m.

**8)** How fast is the ball moving just before it lands?

**Answer: (d) 27.1 m/s**

Energy: $v = \sqrt{(v_0^2 + 2gh)} = \sqrt{(144 + 588.6)} = 27.1$ m/s.

**9)** If the ball were launched at 30° below the horizontal with the same speed, its speed just before landing would be

**Answer: (a) the same.**

$v = \sqrt{(v_0^2 + 2gh)}$ depends only on the speed and height, not the direction.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

An airplane flies at 250 km/h relative to the air. A wind blows toward the east at 60 km/h.

**10)** The pilot points the plane due north. What is its speed relative to the ground?

**Answer: (c) 257 km/h**

$v = \sqrt{(250^2 + 60^2)} = 257$ km/h.

**11)** To travel due north relative to the ground, the pilot should point the plane at what angle west of north?

**Answer: (d) 13.9 °**

$\sin\theta = 60/250$, $\theta = 13.9°$.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A 70 kg student stands on a bathroom scale in an elevator.

**12)** The elevator accelerates upward at 1.8 m/s². What does the scale read?

**Answer: (a) 813 N**

$N - mg = ma$, so $N = 70(9.81 + 1.8) = 813$ N.

**13)** Later the elevator is moving downward and slowing down. The scale reading is

**Answer: (c) greater than mg.**

Slowing while moving down means the acceleration points up, so $N > mg$.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A horizontal force of 36 N pushes a 5 kg block, which pushes a 7 kg block in front of it across a frictionless floor.

**14)** What is the magnitude of the force the 5 kg block exerts on the 7 kg block?

**Answer: (b) 21 N**

$a = F/(m_1 + m_2) = 3$ m/s²; the front block needs $F = (7)(3) = 21$ N.

**15)** If the same force pushed on the 7 kg block instead, the contact force between the blocks would be

**Answer: (b) smaller.**

Now the contact force only accelerates the 5 kg block: $(5)(3) = 15$ N.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A 3 kg block slides down an incline at 35°. The coefficient of kinetic friction is 0.25.

**16)** What is the acceleration of the block?

**Answer: (b) 3.62 m/s²**

$a = g(\sin 35° - \mu_k \cos 35°) = 3.62$ m/s².

**17)** If the block were instead sliding up the incline, the magnitude of its acceleration would be

**Answer: (b) larger.**

Going up, gravity and friction both point down the slope: $a = g(\sin\theta + \mu_k \cos\theta)$.

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**THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.**

A car rounds a flat (unbanked) curve of radius 50 m. The coefficient of static friction between the tires and the road is 0.6.

**18)** What is the maximum speed at which the car can round the curve?

**Answer: (b) 17.2 m/s**

$\mu_s mg = mv^2/r$, so $v = \sqrt{(\mu_s g r)} = 17.2$ m/s.

**19)** A heavier car on the same curve would have a maximum speed that is

**Answer: (c) the same.**

The mass cancels in $\mu_s mg = mv^2/r$.

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**THE NEXT QUESTION PERTAINS TO THE SITUATION DESCRIBED BELOW.**

A planet has half the radius of Earth and one eighth of Earth's mass. An astronaut weighs 600 N on Earth.

**20)** What does the astronaut weigh on the planet?

**Answer: (d) 300 N**

$g \propto M/R^2 = (1/8)/(1/2)^2 = 1/2$, so $W = 300$ N.

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